

Project Proposal: Automated Smart Lock Arm System

1. Company Overview

Amazon's logistics operations are divided into three main areas: First Mile, Middle Mile, and Last Mile.

- First Mile: Involves receiving customer orders at fulfillment centers, picking items from shelves, packaging them, and loading them onto vehicles.
- Middle Mile: Packages move to sortation centers, are sorted by region, and loaded onto trucks for distribution.
- Last Mile: Packages are sorted by postcode at delivery stations and delivered to customers' homes.

Transport Operations Management (TOM) handles the movement of lorries and manages the warehouse yards. Yards are the busiest and most dangerous areas due to high volumes of trailer movements. The TOM Engineering team is tasked with improving safety in these yards.

2. Project Overview

The goal is to design an engineering solution to automate the current manual Stop and Lock system used during the trailer docking process. This will enhance safety by reducing human involvement in close proximity to moving lorries.

The existing process requires a yard marshal to manually place a Stop and Lock barrier in front of a lorry after it docks. This is a high-risk task, especially during peak periods with up to 250 trailer movements daily.

3. Proposed Solution – Smart Lock Arm System

We propose an automated Smart Lock Arm System, a mechanical safety arm equipped with sensors that deploys in front of a docked lorry to prevent premature movement. It works as follows:

1. Lorry parks at the dock.
2. Indoor team confirms docking via app.
3. Arm automatically swings out to block the lorry and outdoor light turns red.
4. Unloading begins.
5. When finished, team confirms via app, the arm retracts, and the light turns green.
6. Driver checks signal and departs safely.

This arm will be connected to Amazon's existing red/green light and combi lock systems and operate in all weather conditions.

4. Business Rationale

- Enhances yard safety by eliminating the need for manual barrier placement.
- Reduces accident risks during high-traffic periods.
- Ensures smoother operation and saves time.
- Compatible with existing systems and infrastructure.
- Cost-effective and scalable across European Amazon sites.

5. Pilot and Rollout Timeline

- Design & Planning: 2 weeks
- Build Prototype: 3 weeks
- Pilot Testing: 4 weeks
- Feedback & Adjustments: 2 weeks
- Final Approval: 1 week
- Europe Rollout: After Christmas peak (Jan onwards)

6. Team Work Distribution

- Member 1 – Technical Designer: Design the mechanical safety arm and app integration.
- Member 2 – Risk and Feasibility Analyst: Evaluate safety, weather compatibility, and cost.
- Member 3 – Report Writer: Compile the report with all technical and strategic information.
- Member 4 – Presentation Lead: Prepare slides and plan the pilot timeline.